Publication List – Sean Patrick MacBride

Publication list current as of October 24, 2026.

1. Rodeghiero, G., Drlica-Wagner, A., Taranto, A., Rosignoli, L., Pollek, H., Pai, A., Jones, L., Howard, E., MacBride, S., Andrew, J., *et al.* An overview of stray light findings and interpretation during on-sky commissioning of LSSTCam. *arXiv preprint arXiv:2606.31945* (2026).
2. Abac, A., North, C., Bhandare, R., Pierra, G., Amato, A., Chen, D., Haskell, B., Robinet, F., Fyffe, M., Muccillo, L., *et al. arXiv: GWTC-5.0: Methods for Identifying and Characterizing Gravitational-wave Transients* tech. rep. (2026).
3. Wüthrich, J., Zhao, G., Yamani, B., Lebrun, L., MacBride, S., Fazan, A., Andrade-Oliveira, F. & Soares-Santos, M. Characterizing robotic positioners under the influence of changing gravity vectors for future spectroscopic surveys. *arXiv preprint arXiv:2606.26821* (2026).
4. MacBride, S. P., McMahon, I., Laghi, D., Sánchez-Cid, D., Andrade-Oliveira, F., Soares-Santos, M., Bilicki, M., Ferraiuolo, S., More, S., Gray, R., *et al.* Constraints on the Cosmic Expansion Rate Using the DECam Local Volume Exploration Survey and the Fifth Gravitational Wave Transient Catalog. *arXiv preprint arXiv:2610.25909* (2026).
5. Abac, A., Abe, A., Abouelfettouh, I., Acernese, F., Ackley, K., Adam, A., Adhicary, S., Adhikari, D., Adhikari, R., Adkins, V., *et al.* Constraints on ultralight bosons from merging binary and remnant black holes observed during the second and third parts of the fourth LIGO-Virgo-KAGRA observing run. *arXiv preprint arXiv:2608.11620* (2026).
6. Furesz, G., Simcoe, R., Angile, E., Molina, G. B., Egan, M., Gabutti, M., Lazar, J., Malonis, A., Mastersson, R., Mitchell, B., *et al.* First year of LLAMAS the large lenslet array Magellan spectrograph: design, build, commissioning, first results, and lessons learned. In *Ground-based and Airborne Instrumentation for Astronomy XI* **14149,** 141491L (2026).
7. Collaboration, L. S., Collaboration, V. & Collaboration, K. GWTC-5.0: Constraints on the Cosmic Expansion Rate and Modified Gravitational-wave Propagation. *arXiv preprint arXiv:2605.27227* (2026).
8. Abac, A., Abe, A., Abouelfettouh, I., Acernese, F., Ackley, K., Adam, A., Adhicary, S., Adhikari, D., Adhikari, R., Adkins, V., *et al.* GWTC-5.0: Observations from the Second Part of the Fourth LIGO-Virgo-KAGRA Observing Run and Updates to the Gravitational-Wave Transient Catalog. *arXiv preprint arXiv:2605.27225* (2026).
9. Scientific, L., Van den Bossche, E., *et al.* GWTC-5.0: Observations from the second part of the fourth LIGO-Virgo-KAGRA observing run and updates to the Gravitational-Wave Transient Catalog. *Astrophysical Journal Letters* **1004** (2026).
10. Collaboration, L. S., Collaboration, V., Collaboration, K., *et al.* GWTC-5.0: Observations from the second part of the fourth LIGO-Virgo-KAGRA observing run and updates to the Gravitational-Wave Transient Catalog. *arXiv e-prints,* arXiv–2605 (2026).
11. Broughton, A., Ferté, A., Waters, C. Z., Rykoff, E. S., MacBride, S. P., Malagón, A. A. P., Lupton, R. H. & Urbach, E. K. Image Calibration and Instrument Signal Removal for the First Year of the LSST (2026).
12. Sun, L., Kuns, K., Slagmolen, B., Fritschel, P., Schmidt, P., Lantz, B., Chua, S., Ballmer, S., Barton, M., Cumming, A., *et al.* LIGO A: Detector Design and Science Prospects Beyond A+. *arXiv preprint arXiv:2608.11673* (2026).
13. Fanning, K., Antilogus, P., Aubourg, E., Banovetz, J. D., Bernard, A., Bond, T., Boucaud, A., Bowdish, B., Bregeon, J., Breugnon, P., *et al.* LSSTCam installation and commissioning at the Vera C. Rubin Observatory. In *Ground-based and Airborne Telescopes XI* **14147,** 406–418 (2026).

14. McMahon, I., Laghi, D., Soares-Santos, M., Ackley, K., Dálya, G., Gençel, Y., Sánchez-Cid, D., Andrade-Oliveira, F., MacBride, S., Chapman-Bird, C., *et al.* Measurement of the Hubble constant using the Dark Energy Survey Year 6 Gold galaxy catalog and the fourth Gravitational-Wave Transient Catalog. *arXiv preprint arXiv:2602.04766* (2026).

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17. collaboration, L. S., collaboration, V., collaboration, K., *et al.* Open Data from LIGO, Virgo, and KAGRA through the Second Part of the Fourth Observing Run. *arXiv e-prints,* arXiv–2605 (2026).

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19. Lebrun, L., Wüthrich, J., Fazan, A., MacBride, S., Andrade-Oliveira, F. & Soares-Santos, M. Precision measurements of the axial drift in fiber positioners for spectroscopic surveys using defocus blur. In *Advances in Optical and Mechanical Technologies for Telescopes and Instrumentation VII* **14154,** 1785–1796 (2026).

20. Broughton, A., Ferté, A., Waters, C. Z., Rykoff, E. S., MacBride, S. P., Plazas Malagón, A. A., Lupton, R. H. & Urbach, E. K. *RTN-117: Image Calibration and Instrument Signal Removal for the First Year of the LSST* tech. rep. (NSF-DOE Vera C. Rubin Observatory, 2026).

21. Jones, R. L., Yoachim, P., Neilsen Jr, E. H., Ribeiro, T., Dennihy, E. & MacBride, S. P. Scheduling the LSST: commissioning the Rubin feature-based scheduler. In *Observatory Operations: Strategies, Processes, and Systems XI* **14151,** 378–394 (2026).

22. Abac, A., Abe, A., Abouelfettouh, I., Acernese, F., Ackley, K., Adam, A., Adhicary, S., Adhikari, D., Adhikari, R., Adkins, V., *et al.* Search for Compact Binary Coalescences with Sub-solar Mass Components in Data from the Second Part of LIGO-Virgo-KAGRA's Fourth Observing Run. *arXiv preprint arXiv:2610.20276* (2026).

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25. MacBride, S. P., Roodman, A., Marshall, S., Utsumi, Y., Fanning, K., Banovetz, J., Schutt, T., Broughton, A., Liang, S., Rasmussen, A. P., *et al.* The on-sky performance of the LSST Camera CCD array. In *X-Ray, Optical, and Infrared Detectors for Astronomy XII* **14157,** 272–292 (2026).

26. MacBride, S. P., Jones, R. L., Yoachim, P., Ribeiro, T., Guy, L. P., Anand, S., Howard, E. L., Sullivan, I. S., Wang, D. L., Bellm, E. C., *et al.* The Rubin Observatory target-of-opportunity system in the first year of operations. In *Observatory Operations: Strategies, Processes, and Systems XI* **14151,** 326–346 (2026).

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28. Juramy, C., Antilogus, P., Astier, P., Banovetz, J., MacBride, S. P., Rasmussen, A. P. & Utsumi, Y. The" dark dips" phenomenon in the LSST Camera on-sky images. *arXiv preprint arXiv:2607.00925* (2026).

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30. Abac, A., Abe, A., Abouelfettouh, I., Acernese, F., Ackley, K., Adam, A., Adamcewicz, C., Adhicary, S., Adhikari, D., Adhikari, R., *et al.* Updated Upper Limits on the Isotropic Gravitational-Wave Background from LIGO, Virgo, and KAGRA Data through April 2025. *arXiv preprint arXiv:2608.23477* (2026).

31. Zhang, H., Kokubo, M., MacBride, S., McMahon, I., Tominaga, N., Utsumi, Y., Yoshida, M., Morokuma, T., Tanaka, M., Arai, A., *et al.* A Joint Search for the Electromagnetic Counterpart to the Gravitational-Wave Binary Black-Hole Merger Candidate S250328ae with the Dark Energy Camera and the Prime Focus Spectrograph. *arXiv preprint arXiv:2508.00291* (2025).

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33. Team, L. C., Antilogus, P., Astier, P., Banovetz, J., Bregeon, J., Chiang, J., Digel, S. W., Fanning, K., Faris, Y., Guillemin, T., *et al.* LSST Camera Electro-Optical Test (Run 7) Results. *Commissioning Technical Note SITCOMTN-148, NSF-DOE Vera C. Rubin Observatory (August 2025)* (2025).

34. AlSayyad, Y., Axelrod, T. S., Banovetz, J., Bechtol, K., Bellm, E. C., Bosch, J. F., Bradshaw, A., Broughton, A., Budlong, A., Caplar, N., *et al.* PSTN-019: The LSST Science Pipelines Software: Optical Survey Pipeline Reduction and Analysis Environment. *NSF-DOE Vera C. Rubin Observatory Technical Report,* 32 (2025).

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36. Chandler, C., Kurlander, J., Bernardinelli, P., Juri, M., Eggl, S., Singh, D., Schwamb, M. E., Hsieh, H., Sullivan, I., Jones, R. L., *et al.* Solar System Science with the NSF-DOE Vera C. Rubin Observatory: Overview and Results. In *AGU Fall Meeting Abstracts* **2025,** P32B–05 (2025).

37. Borghi, N., Moresco, M., Anderson, R. I., Carbone, C., Cimatti, A., Escoffier, S., Giocoli, C., MacBride, S., Majidi, F. Z., Milakovi, D., *et al.* Standard Sirens in 2040s: Probing the Cosmic Expansion History with Gravitational Waves and Spectroscopic Galaxy Surveys. *arXiv preprint arXiv:2512.18369* (2025).

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